Samuel Underwood

Sensing the World Reflection:

Over the course of the first two weeks here at Lehigh, we have done many things in our Engineering 095 section. The first day was an overview of the course including how to get in contact with professors, the late work policy, descriptions of what the engineering disciplines involved do, and some of the materials that each use. In our second class we learned about the Wilbur and Whitaker labs and the different resources available at Wilbur. We also started to experiment with different materials trying to find their resistance, as well as briefly talking about sensing materials. Then the third day in Whitaker we worked to make electrodes out of the materials we were provided with including Ag paste, Ag paint, and carbon paint. We did this by cutting shapes out of tape to create a basic electrode and then filled in the area with the material of our choosing. We also explored the cleanroom here on campus and the process of MOCVD as well as photolithography. On day 4 we created paper circuits in the shape of a butterfly attempting to get 4 small LEDs to light up. We used copper tape to create the circuit, put a battery in the middle, and then used copper tape on the antennas to complete the circuit and activate the LEDs. We worked in Whitaker to finish some sensors by putting a PEDOT-PSS coating on the electrodes we made previously. This was done by using a spin coater to evenly spread the coating across our electrodes on our glass slides. Finally, on day 5 we worked in the classroom learning more about programming, the programming approach, flowcharts, setting up a GitHub account and creating a portfolio that I can utilize in the future.

I learned loads and gained experience over the first few weeks of this course. In the beginning, my group struggled to understand the formula for measuring resistivity. We weren't exactly sure how to even make the electrodes. However, after trial and error and learning from our mistakes, we were able to figure out how to calculate the resistivity correctly and how to properly make the electrodes so that we take accurate measurements. We tried to split up the work so that someone was working on the beginning part of the sensor fabrication, which was laying down the shape. Then we had someone else working on applying the material and heating it. Finally, we had someone doing the final touches and using the spin coating to apply the PEDOT-PSS. Learning about circuits was also interesting and beneficial as I have little to no background in electronics or electrical engineering. It was a fun challenge to try and create this circuit. The hardest part for me was making sure that my connection was strong throughout, as there were many places where I did not overlay the tape well, creating a weak current. This then caused the LED connected to that portion of the tape to not function properly. Overall, I have learned a lot and am excited to continue with this module.